

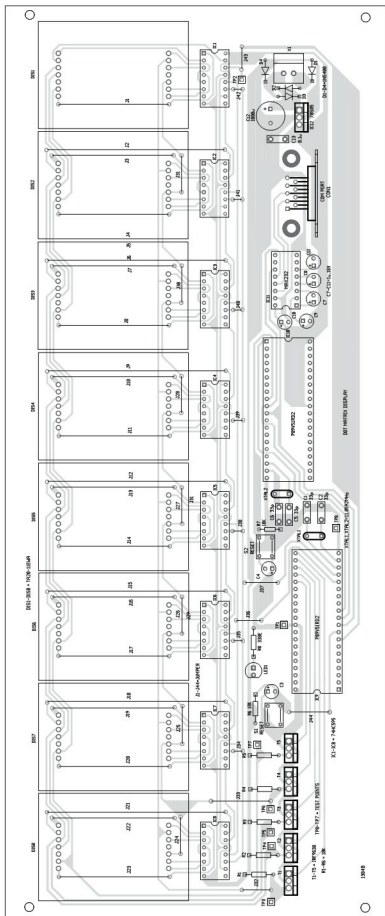


Fig. 5: Component layout for the PCB (figure scaled down to 74 per cent)

for cascading. Output enable ($\overline{OE}$) pin is connected to ground for data in the storage register to appear at the output automatically. Microcontroller port pins P2.0 through P2.4 are connected to gates of MOSFETs T1 through T5 to drive columns C1 through C5 of all the dot-matrix display modules.

Resistor R6 and capacitor C3 provide the power-on reset to IC9. Switch S1 is used for manual reset. An 11.0592MHz ($X_{T,AL}$) crystal along with two 33pF capacitors provides the basic clock frequency to microcontroller IC9.

Working of the system is straightforward. When you power-on the system after complete set-up, "No Msg." is displayed intimating that the system is waiting for a new message. Write an eight-character SMS with a '#' (which is the ninth character) at the end to indicate completion of the message. Send the message to the mobile number of the SIM in the GSM module. The same message will be shown immediately on the display.

Construction and testing

PCB layout for the moving-message display is shown in Fig. 4 and its component layout in Fig. 5. Please note that both these figures are scaled down to fit in the page. The actual-size PCB pattern is included in the DVD and can be downloaded from www.efymag.com. Assemble the circuit on a PCB as it saves time and minimises assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC bases for ICs. Before inserting the ICs, check the supply voltage.

First, insert the SIM in the GSM module and power it 'on.' Red and green LEDs of the GSM module indicate power-on and signal reception, respectively. The GSM module is connected to the circuit through CON1. Make a call on the SIM number in the module. A ring will be heard if the call gets connected successfully. Now, send the message through your mobile. It will immediately appear on the display.

Check the correct power supply at TP1 with respect to TP0. You can ob-